## Phase 2 – Proposal and Code Implementation

# Deforestation Monitoring and Land-Cover Classification Using Deep Learning and Transfer Learning


### Submitted To

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## 1. Project Title

Deforestation Monitoring and Land-Cover Classification Using Deep Learning and Transfer Learning

## 2. Background and Motivation

Environmental monitoring through satellite imagery plays a significant role in identifying land-cover changes and deforestation activities. Manual interpretation of satellite images is time-consuming and difficult to scale. Deep Learning techniques, especially Convolutional Neural Networks (CNNs), provide an automated solution for analyzing large-scale satellite imagery and classifying environmental land-cover categories.

This project investigates the effectiveness of CNN-based and Transfer Learning approaches for satellite image classification using the Planets Dataset.

## 3. Problem Statement

The objective of this project is to develop an automated image classification system capable of identifying different land-cover categories from satellite images. The project compares a Custom CNN model with Transfer Learning architectures to determine the most effective approach for environmental monitoring applications.

## 4. Selected Dataset

Dataset Name: Planets Dataset

Dataset Link: https://www.kaggle.com/datasets/nikitarom/planets-dataset

## 5. Dataset Description

The Planets Dataset contains satellite images representing various environmental and land-cover conditions.

### Dataset Characteristics

- Number of Images: 40,479
- Number of Classes: 17
- Image Type: Satellite Images
- Classification Type: Multi-Label Image Classification

### Example Classes

- Agriculture
- Artisinal Mine
- Bare Ground
- Blooming
- Blow Down
- Primary Forest
- Water
- Road
- Haze

The dataset contains class imbalance where certain categories appear less frequently than others.

## 6. Research Questions

1. Can deep learning models accurately classify satellite imagery for environmental monitoring?
2. How does a Custom CNN compare with Transfer Learning models?
3. Which model achieves the highest classification performance?
4. Does Transfer Learning improve classification accuracy and reduce prediction errors?

## 7. Expected Results

The expected outcome is a robust image classification framework capable of identifying environmental land-cover categories from satellite imagery.

Expected findings include:

- High classification accuracy across multiple land-cover classes.
- Better performance from Transfer Learning models compared to a Custom CNN.
- Effective automated environmental monitoring capability.

## 8. Proposed Methodology

The project follows the workflow below:

1. Dataset Loading
2. Data Preprocessing
3. Image Resizing
4. Image Normalization
5. Data Augmentation
6. Train-Validation-Test Split
7. Custom CNN Development
8. EfficientNetB0 Development
9. ResNet50 Development
10. Model Training
11. Model Evaluation
12. Performance Comparison
13. Error Analysis

## 9. CNN Model Design

The Custom CNN architecture consists of:

- Convolutional Layers
- Max Pooling Layers
- Dropout Layers
- Global Average Pooling
- Dense Layers
- Sigmoid Output Layer

The model is designed for multi-label image classification.

## 10. Transfer Learning Models

### EfficientNetB0

EfficientNetB0 utilizes pretrained ImageNet weights and provides efficient feature extraction while maintaining high classification performance.

### ResNet50

ResNet50 utilizes residual learning connections, enabling deeper feature extraction and improved classification capability.

## 11. Evaluation Metrics

The following metrics were used:

- Binary Accuracy
- Loss
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Classification Report
- Model Comparison Analysis

## 12. Expected Figures and Tables

### Figures

- Dataset Class Distribution
- Accuracy Curves
- Loss Curves
- ROC Curves
- Model Accuracy Comparison
- Model Loss Comparison
- Performance Heatmap

### Tables

- Dataset Summary Table
- Classification Reports
- Model Performance Comparison
- Error Analysis Table
- Final Conclusion Table

## 13. Implementation Summary

Five Kaggle notebooks were developed:

### Notebook 1 – Dataset Analysis and Preprocessing

https://www.kaggle.com/code/sayankrdutta/01-dataset-analysis-and-preprocessing

### Notebook 2 – Custom CNN Model

https://www.kaggle.com/code/sayankrdutta/02-custom-cnn-model

### Notebook 3 – EfficientNetB0 Deforestation Monitoring

https://www.kaggle.com/code/sayankrdutta/03-efficientnetb0-deforestation-monitoring

### Notebook 4 – ResNet50 Deforestation Monitoring

https://www.kaggle.com/code/sayankrdutta/04-resnet50-deforestation-monitoring

### Notebook 5 – Model Comparison, GradCAM and Error Analysis

https://www.kaggle.com/code/sayankrdutta/05-model-comparison-gradcam-error-analysis

## 14. Results

| Model | Binary Accuracy | Loss |
|---|---|---|
| Custom CNN | 95.18% | 0.1217 |
| EfficientNetB0 | 96.24% | 0.0977 |
| ResNet50 | 95.93% | 0.1062 |

### Best Performing Model

EfficientNetB0 achieved the highest classification accuracy and lowest loss among all evaluated models.

## 15. GitHub Repository

Project Repository:

https://github.com/Sayankrdutta07/42371949_Dutta_Sayan-Kr-Dutta_Machine-Learning-project-Phase-2

The repository contains:

- Source Code
- Kaggle Notebooks
- Dataset Information
- README File
- Output Figures
- Project Documentation

## 16. Conclusion

This project demonstrates the effectiveness of Deep Learning and Transfer Learning techniques for satellite image classification and environmental monitoring. Experimental results indicate that Transfer Learning approaches outperform a Custom CNN architecture, with EfficientNetB0 achieving the highest overall performance.

The developed framework can support automated land-cover classification and environmental monitoring applications using satellite imagery.

## References


1. Planets Dataset – Kaggle
2. TensorFlow Documentation
3. Keras Documentation
4. EfficientNet Research Paper
5. ResNet Research Paper